Emotion Recognition from Speech

A Machine Learning Approach to Acoustic Affective Computing

COMP-SCI 5530: Principles of Data Science | University of Missouri - Kansas City

The Research Team

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The Problem: Decoding Affect

- Speech is the primary medium for human self-expression, carrying rich paralinguistic cues.
- Traditional HCI (Human-Computer Interaction) lacks emotional intelligence.
- Goal: Build a robust pipeline to extract emotional states from raw voice waves.
- Challenges: Subjective nature of emotion and acoustic frequency overlap.

The Foundation: RAVDESS Corpus

- Ryerson Audio-Visual Database of Emotional Speech and Song.
- 24 professional actors (12 Male, 12 Female) ensuring demographic balance.
- Lexically-matched neutral statements to isolate emotion from semantics.
- 8 Fundamental Emotions: Neutral, Calm, Happy, Sad, Angry, Fearful, Disgust, Surprised.

Phase I: Signal Processing & Augmentation

- Feature Extraction: MFCCs, Chroma Features, and Mel Spectrograms.
- Standardization: 16kHz Mono audio with silence truncation.
- Data Augmentation: White noise injection, Pitch shifting, and Time stretching
- Advanced Technique: Speaker-Mix Augmentation to improve generalization.

Phase II: Hybrid Multi-Task Architecture

- Backbone: HUBERT (Hidden-Unit BERT) for deep speech representations.
- Engineering Branch: Handcrafted acoustic descriptors for prosody (Pitch, Energy, ZCR).
- Fusion: Combining contextual transformer embeddings with interpretable features.
- Multi-Task Heads: Simultaneous prediction of Emotion (Class) and Intensity (Degree).

Evaluation: Quantitative Success

- High Arousal Robustness: 80-85% specific accuracy for Happy/Angry classes.
- Intensity Prediction: Reached ~78% accuracy, demonstrating strong degree classification.
- Macro F1 Strategy: Ensuring equal priority across all emotional frequencies.
- Per-Actor Variance: Identifying speaker-independent vs. speaker-dependent traits.

Live Inference: The Streamlit Dashboard

- Real-time Telemetry: Processing microphone streams in rolling windows.
- Edge-Ready: Low-latency inference (avg. ~370ms) for interactive use.
- Visualization: Dynamic probability trackers for 8 emotion classes.
- Challenge: Managing micro-batch noise and instability in live environments.

Ethical AI & Future Responsibility

- Algorithmic Bias: Mitigating demographic skew through diverse dataset integration.
- Privacy & Surveillance: Implementing encrypted, localized on-device processing.
- Informed Consent: Ensuring users are aware of emotional tracking protocols.
- Transparency: Using engineered features to explain the model's decision boundaries.

Conclusion & Future Horizons

- Established a viable end-to-end Affective Computing pipeline.
- Future: Multimodal integration (Computer Vision + Audio) for higher reliability.
- Scale: Moving to the edge for mental health and automotive safety applications.
- Legacy: Setting a high standard for data-driven emotional intelligence.

Thank You!